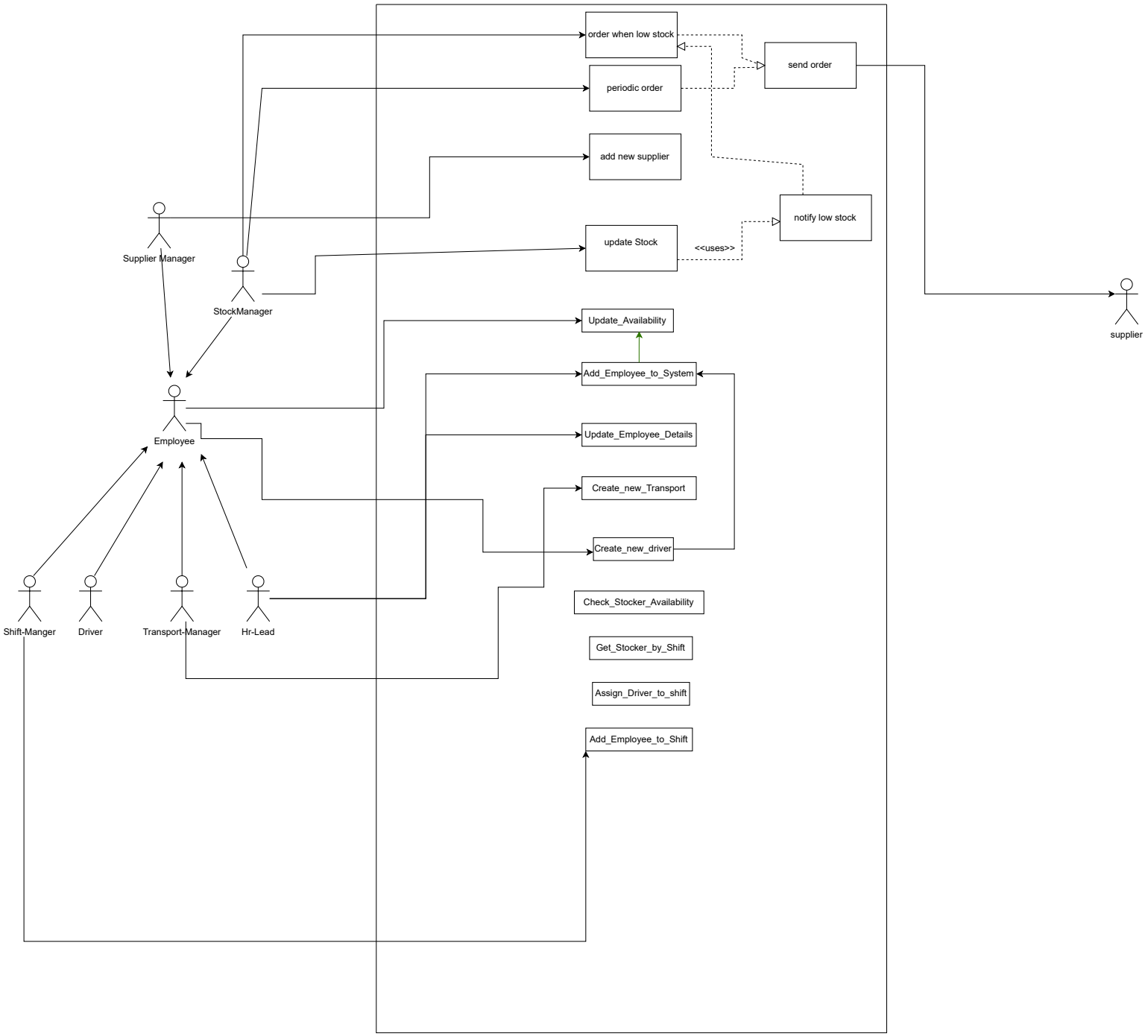
Names and IDs:

Yael Durahly | 212985535
Denis Coleman | 211408356
Tom Gluzman | 322861477
Ethen Krimer | 323888412

use case diagram



:1 use case מימוש

Use case name:
Add employee to shift

Textual Description:
The process begins with the manager providing input to specify the desired shift ID, the employee ID of the worker to be added, and the role the employee will take on within that shift.
If all inputs meet the necessary conditions, the employee is successfully assigned to the shift with the designated role. Otherwise, appropriate error messages are displayed to notify the manager.

List of Actors:
Shift Manager

Pre-conditions:
The shift ID entered must correspond to an existing shift in the ShiftsDB, the employee ID must correspond to an existing employee in the EmployeesDB, the shift being assigned must have a designated Shift Manager, the employee must be available for the specific type of shift being worked with, the employee must possess the required role for the shift.

Post-conditions:
a copy of the specified employee object is added to the assigned employees list within the corresponding shift object.

Main success scenario:

PRINT "Enter the Employee ID you want to assign to a shift:"
READ employeeID from manager input
employee ← findEmployeeById(employee ID)

PRINT "Enter the Shift ID to assign the employee to:"
READ shiftID from input
shift ← findShiftById(shift ID)

PRINT "Select the corresponding number to the desired Role to add:"
CALL displayAvailableRoles()
selectedRole ← promptForRole(choice)

IF succeeded to add employee to shift THEN
PRINT "Employee <employee name> added to shift <shift ID> with role <roleName>"

Alternatives/Extensions:

In case of a failure – we display the specific reason for failure.

IF employee does not exist THEN
    PRINT "Employee with ID <employee ID> does not exist."
    RETURN

```
IF shift does not exist THEN
    PRINT "Shift with ID <shift ID> does not exist."
    RETURN

isAvailable ← checkEmployeeAvailability(employee, shift.date)
IF NOT isAvailable THEN
PRINT "Employee <employee name> is not available for this shift date."
RETURN

IF employee already assigned to shift THEN
     PRINT "Employee <employee name> is already assigned to this shift."
      RETURN

IF employee NOT has selected role THEN
      PRINT "Employee <employee name> does not have the role <roleName>"
      RETURN

IF NOT succeeded to add employee to shift THEN
     PRINT "Failed to add employee to the shift."
```

: 1 חלק נוסף

contract:

Operation:
addEmployeeToShift()

Cross References:
Use Case: Add employee to shift

Preconditions:

- The specified shift id exists in the ShiftsDB.
- The specified employee id exists in the EmployeesDB.
- The shift has an assigned Shift Manager.
- The employee is available for the type of shift being assigned.
- The employee possesses the required role for the shift.

Postconditions:

- A copy of the specified employee object was added to the assigned employees list within the corresponding shift object.
- A success message is displayed if the employee is successfully added.

:2 use case מימוש

Use case name: Create a transport
Textual Description: The system gets an items list, a branch and a designated delivery date and time. The system creates transport which consists of transport routes from supplier(s) to branch. We attach a truck and a relevant driver.
List of Actors: Transport Manager, Driver.
Pre-conditions: There exists a Stocker at the branch at the given delivery date-time. There exists a suitable supplier for at least part of the required items. There exists a relevant truck for the delivery at the required time. And there exists a driver for the relevant time frame with a suitable license type.
Post-conditions: We create a transport with association to transport routes which associates to a delivery file, the transport also associates with a truck and a driver.
Main success scenario: INPUT: itemsList, branchId, deliveryDateTime 1. VERIFY there is a Stocker available at the branch at deliveryDateTime 2. FIND suppliers who can supply at least part of the items 3. IF no suitable supplier found → FAIL 4. FIND a truck that is available at deliveryDateTime 5. IF no truck found → FAIL 6. FIND a driver who: - is available at deliveryDateTime - has a valid license for the truck 7. IF no suitable driver found → FAIL 8. CREATE Transport entity 9. FOR each supplier: a. CREATE TransportRoute from supplier to branch b. CREATE DeliveryFile for route c. ASSOCIATE DeliveryFile with TransportRoute d. ADD TransportRoute to Transport 10. ASSOCIATE truck and driver with the Transport 11. SAVE Transport to database OUTPUT: Transport created successfully

Alternatives/Extensions:

IF suppliers.isEmpty():

DISPLAY "No supplier found for required items"

ABORT transport creation

IF truck == null:

DISPLAY "No truck available at the required time"

ABORT transport creation

IF driver == null:

DISPLAY "No driver available or valid for selected truck"

ABORT transport creation

IF only some items can be fulfilled:

CREATE transport routes for available items

FLAG transport as "Partially fulfilled"

IF truck or driver already assigned to another transport at same time:

DISPLAY "Truck or driver already in use"

ALLOW user to select different resources or reschedule

חוזה מס' 2:

Contract:

Operation: createTransport(itemsList : List<Item>, branchId : int, deliveryDateTime : datetime)

References:
Use Case: Create a Transport

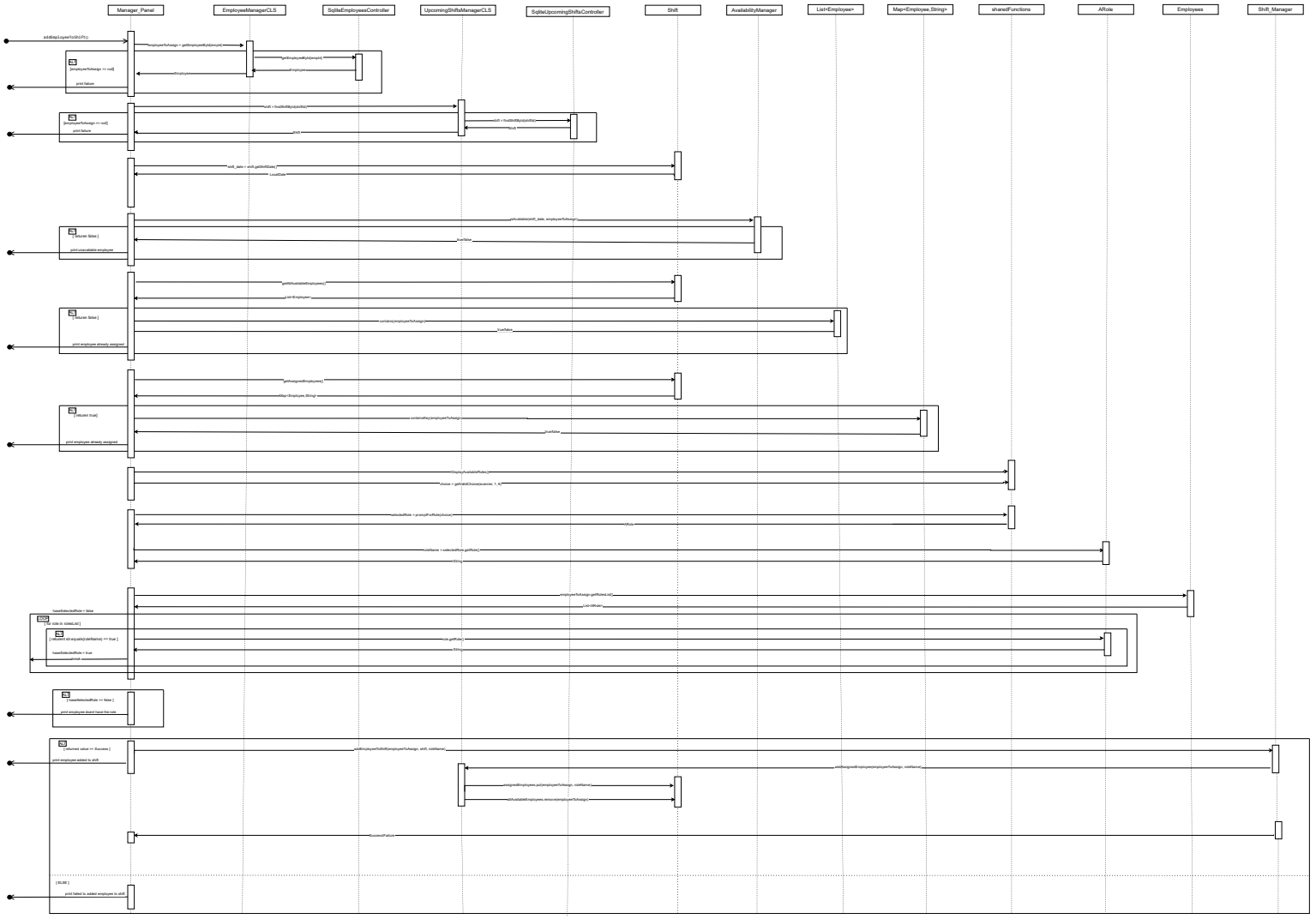
Preconditions:

- There is a Stocker available at the specified branch at deliveryDateTime.
- There exists at least one supplier that can supply part of the requested items.
- There is a truck available for delivery at the required time.
- There is a driver available during the time window with a valid license for the selected truck.

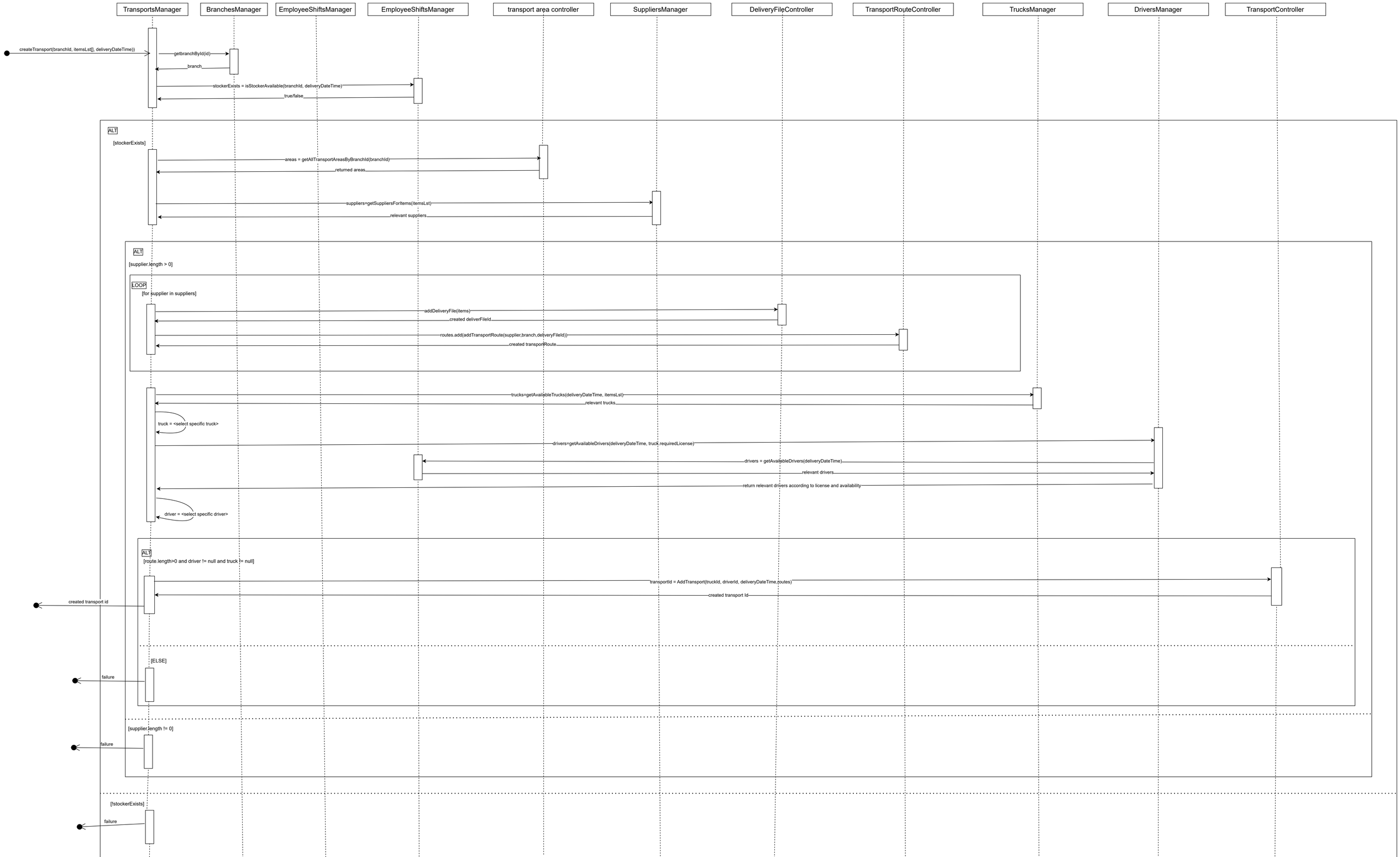
Postconditions:

- A Transport instance was **created**.
- The Transport was **associated** with one or more TransportRoute instances.
- Each TransportRoute was **associated** with a DeliveryFile.
- The Transport was **associated** with a selected Truck.
- The Transport was **associated** with a selected Driver.

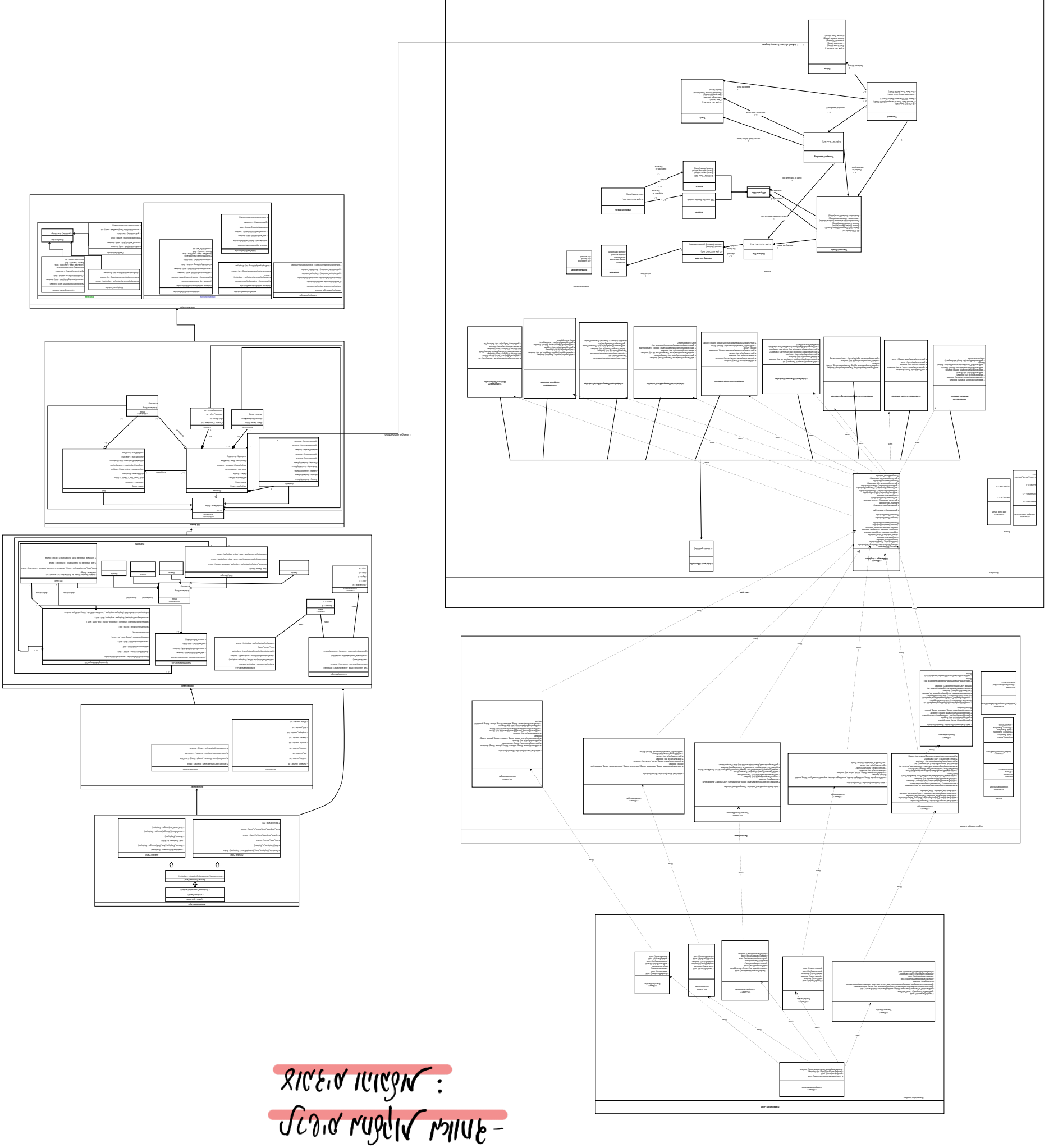
sequence Diagram 1 :



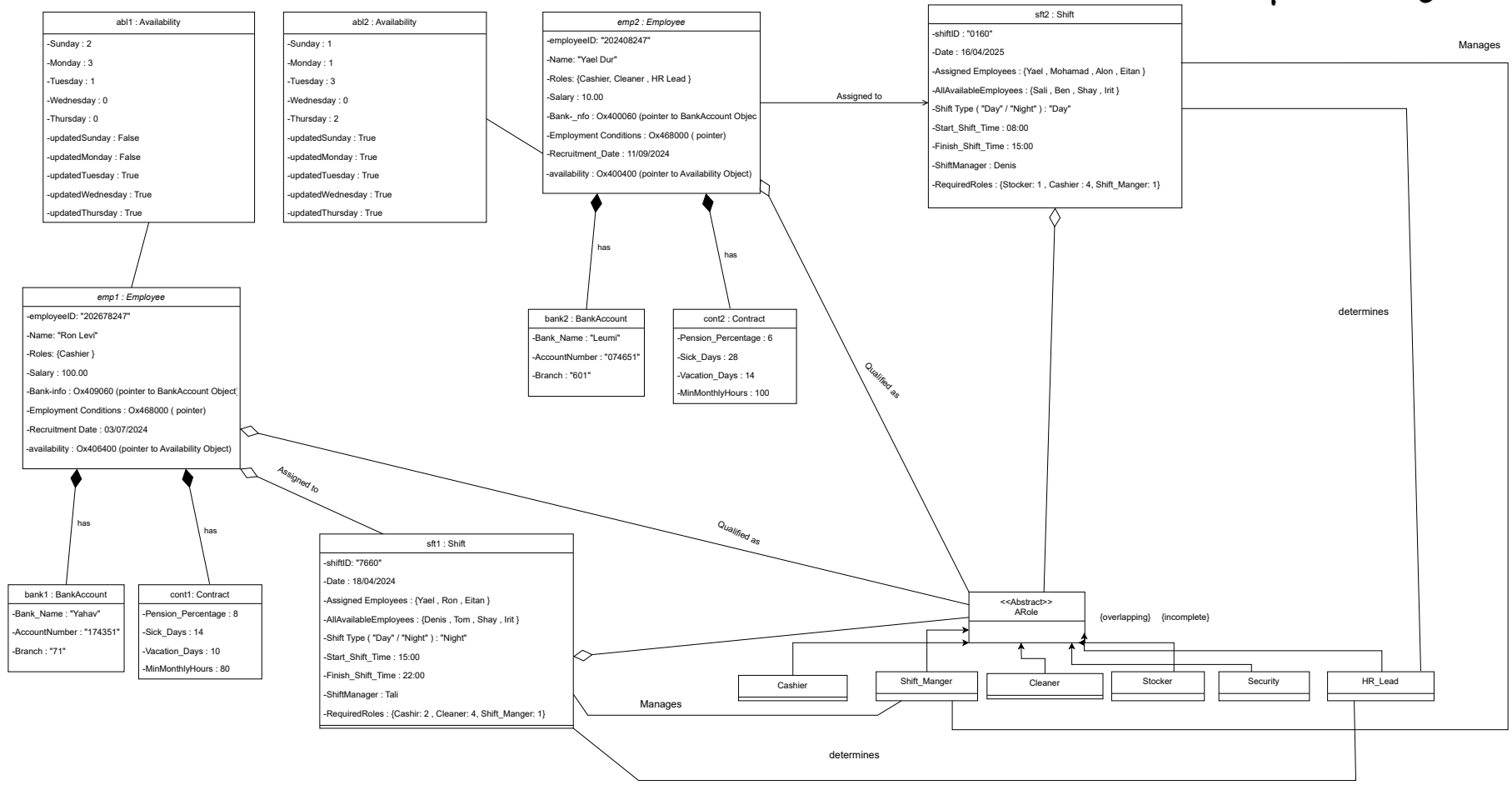
sequence Diagram 2 :


TransportsManager
BranchesManager
EmployeeShiftsManager
EmployeeShiftsManager
transport area controller
SuppliersManager
DeliveryFileController
TransportRouteController
TrucksManager
DriversManager
TransportController
createTransport(branchId, itemsLst[], deliveryDateTime))
getbranchById(id)
branch
stockerExists = isStockerAvailable(branchId, deliveryDateTime)
true/false
ALT
[stockerExists]
areas = getAllTransportAreasByBranchId(branchId)
returned areas
suppliers=getSuppliersForItems(itemsLst)
relevant suppliers
ALT
[supplier.length > 0]
LOOP
[for supplier in suppliers]
addDeliveryFile(items)
created deliverFileId
routes.add(addTransportRoute(supplier,branch,deliveryFileId))
created transportRoute
trucks=getAvailableTrucks(deliveryDateTime, itemsLst)
relevant trucks
truck = <select specific truck>
drivers=getAvailableDrivers(deliveryDateTime, truck.requiredLicense)
drivers = getAvailableDrivers(deliveryDateTime)
relevant drivers
return relevant drivers according to license and availability
driver = <select specific driver>
ALT
[route.length>0 and driver != null and truck != null]
transportId = AddTransport(truckId, driverId, deliveryDateTime,routes)
created transport Id
created transport id
[ELSE]
failure
[supplier.length != 0]
failure
[!stockerExists]
failure

- אנליזה מונחית עצמים בקובץ מלבנית בקובץ :



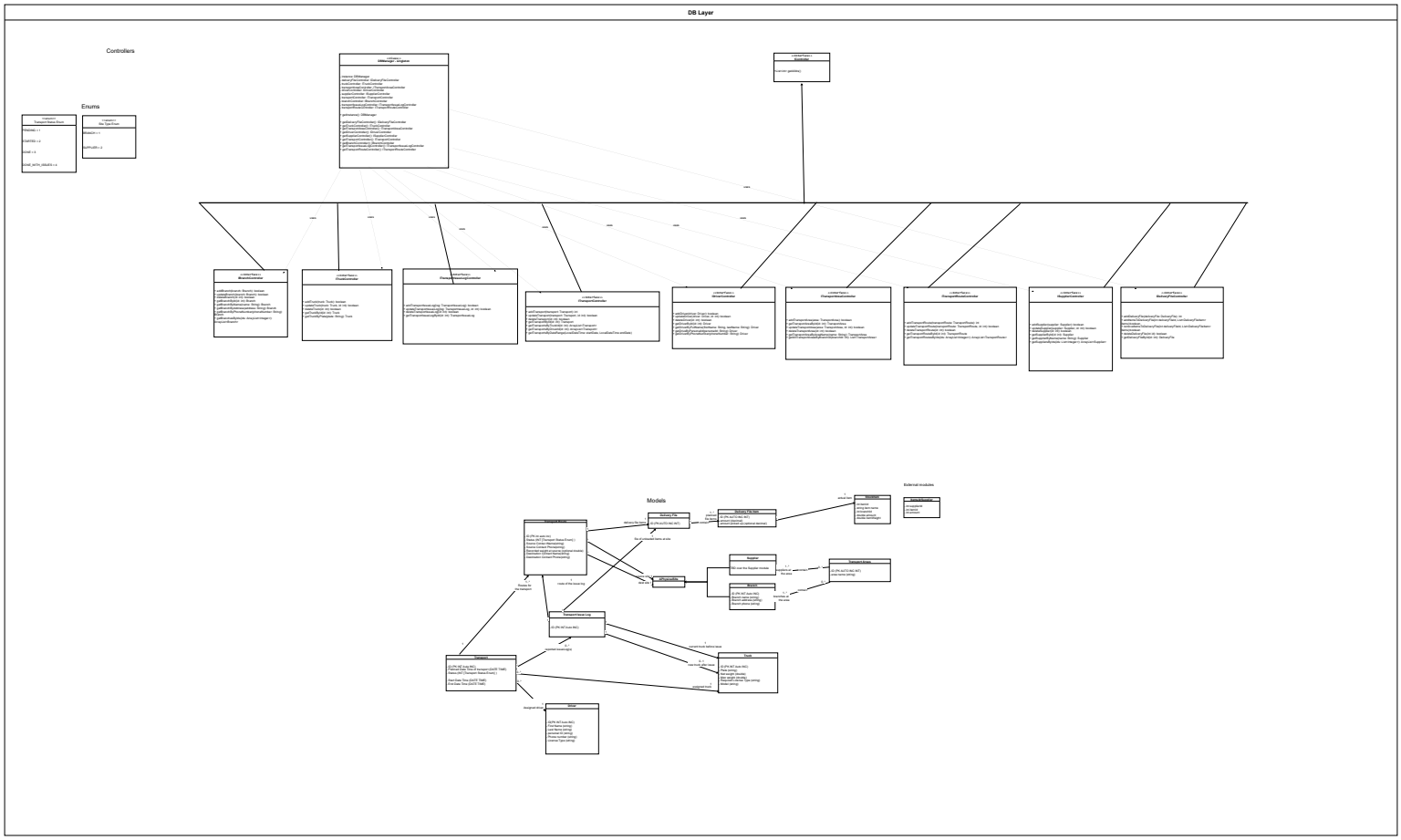
קבלת אובייקטים - אובדים:



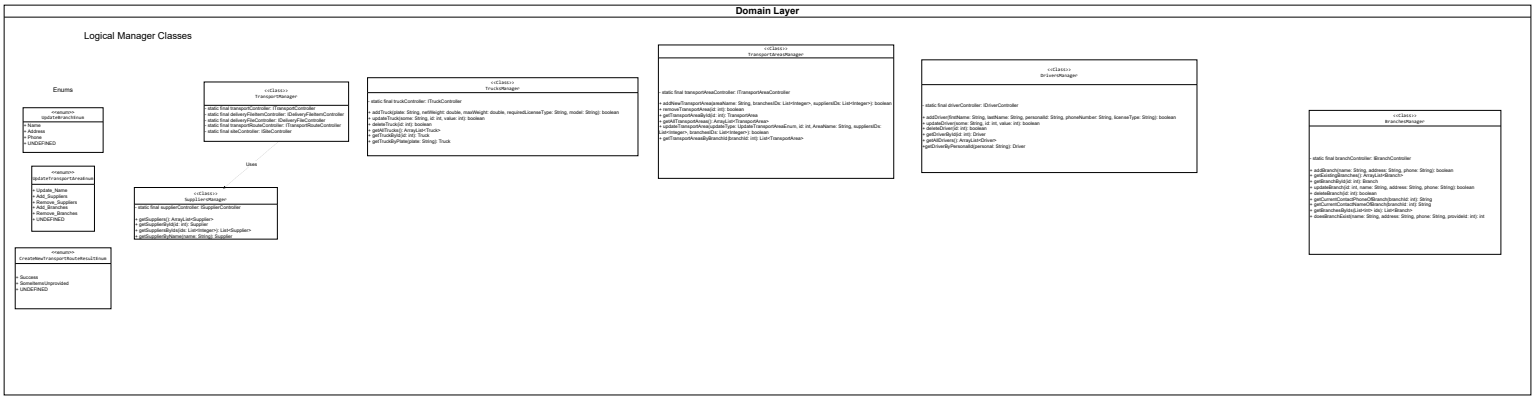
מבנה המערכת – מודלים אדריכלים

* יש לבחור מודל אחד ולפרט את הסיבות לבחירתו.

דיאגרמת DB הנוכחית:



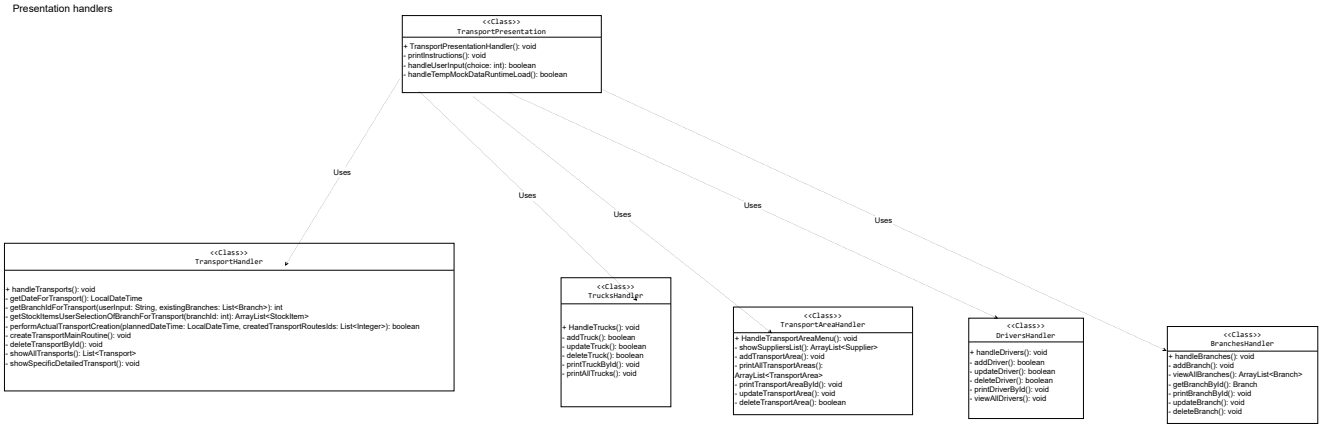
דיאגרמת Domain המלאה:



דיאגרמת המצגת: presentation

Presentation Layer

Presentation handlers



שכפולות עם בדיעות:

ID	Module	Functional/ Non-Functional	Description	Priority	Risk	Status
1.	Workers	F	The system **MUST** support registration of new employees	**MH**	High	Done
2.	Workers	F	The system **MUST** support the termination of an employee	**MH**	High	Done
3.	Workers	F	The system **MUST** contain employees's Name,ID,Bank-info, Salary, employement conditions ,reqruitment date and all role capabilties	**MH**	High	Done
4.	Workers	F	The system **MUST** consider each of employee's schedule restrictions before assigning them into shifts (days + hours)	**MH**	High	Done
5.	Workers	F	The system **MUST** store all employee shift schedule restrictions and preferencesfor scheduling decisions	**MH**	Low	Done
6.	Workers	F	The system **MUST** support to receive employees shifts restrictions until every Thursday.	**MH**	Low	Done
7.	Workers	NF	The system **MUST** contain shifts history for 7 years	**MH**	Low	Done
8.	Workers	F	The system **MUST** allow only users with the role 'Shift Manager' to assign personnel to shifts via the shift management interface.	**MH**	High	Done

9.	Workers	F	The system **MUST** give permission to the HR Lead to update upcoming shift hours	**MH**	Low	Done
10.	Workers	F	The system **SHOULD** give permission to the Shift Manger to update the upcoming shift's emploees even after publishing the upcoming shift	**NTH**	High	Done
11.	Workers	F	The system **SHOULD** warn the manager for committing changes that overstep the boundaries set by the HR lead.	**NTH**	Low	Done
12.	Workers	F	The system **MUST** let only to each Shift Manger to cancel items	**MH**	Low	Done
13.	Workers	NF	The system **MUST** prohibit inserting unqualified personnel into roles for which they lack the required training	**MH**	Low	Done
14.	Workers	F	The system **MUST** confirm that each shift will have ONLY one Shift Manger and the designated ammount of perssonel by the HR	**MH**	Low	Done
15.	Workers	NF	The system **MUST** distinguish between different role logins by ID	**MH**	Low	Done
16.	Workers	F	The system **SHOULD** allow real-time notifications when an employee is assigned or removed from a shift	**NTH**	High	Wont be implmented

17.	Workers	F	The system **SHOULD** allow employees to request shift swaps with other employees through the platform	**NTH**	High	Wont be implmented
18.	Workers	F	The system **SHOULD** send automatic reminders to employees about upcoming shifts and changes	**NTH**	Low	Wont be implmented
19.	Workers	NF	The system **SHOULD** support multilingual UI (Hebrew and English) for ease of use by diverse staff	**NTH**	High	Wont be implmented
20.	Workers	F	The system **SHOULD** allow exporting shift reports/history to PDF	**NTH**	Low	Wont be implmented
21.	Workers & Transportaion	NF	The system **MUST** support scalability to allow the addition of multiple branches.	**MH**	Low	TBD
22.	Workers & Transportaion	F	The system **MUST** support sqLite queries for dataBase usage	**MH**	High	TBD
23.	Workers & Transportaion	F	The system **MUST** ensure ensure drivers assigned to a transport route have a matching shift in the Employees system.	**MH**	High	TBD
24.	Workers & Transportaion	F	The system **MUST** Use consistent Employee.id across both modules (when Driver is an Employee) to avoid desynchronization or duplicate references.	**MH**	Low	TBD

25.	Workers & Transportaion	F	The system **MUST** Validate that any employee assigned to a transportation route has the "Driver" role	**MH**	Low	TBD
26.	Workers & Transportaion	F	Upon creating a new transport route, The system **SHOULD** auto suggest or pre-fill a shift for the driver in the Employees DB, including expected time and branch.	**NTH**	High	TBD
27.	Workers & Transportaion	F	The system **SHOULD** log or alert if a transport driver is double-booked or has overlapping shifts/routes	**NTH**	High	TBD

ID	Module	Functional / Non-Functional	Description	Priority	Risk	Status	
1	Transport	NF	The System must follow this format for Delivery File Item: - ID - Item ID(FK) - amount - amout picked up	MH	High	DONE	
2	Transport	NF	The System must follow this format for Delivery File: - ID - list of Delivery File Item(s) IDs(FKs)	MH	High	DONE	
3	Transport	NF	The System must follow this format for Truck: - ID(Auto INC) - Plate - Neto weight - max weight - Required License Type - Model	MH	High	DONE	

4	Transport	NF	The System must follow this format for Driver: - ID(Auto INC) - First Name - Last Name - personal ID - Phone number - License Type	MH	High	DONE	
5	Transport	NF	The System must follow this format for Branch: - ID - Branch name - Branch address - Branch phone	MH	High	DONE	
6	Transport	NF	The System must follow this format for APhysicalSite: - ID - Site name - Site address - Site phone Branch inheritance from that	MH	High	DONE	

7	Transport	NF	The System must follow this format for Transport Issue Log: - Current Truck ID - New Truck ID - Unloaded items at Site ID - Unloaded Items Delivery File ID - Transport route ID	MH	High	DONE	
8	Transport	NF	For any Transport in the System, a Transport should have: - Transport ID - planned datetime of transport - status (pending, started, done, done with issues) - Truck ID (FK to link to the actual truck) - Driver ID (FK to link to the actual driver) - source - List of APhysicalSite(s) ID(s) (FKs) - destination - List of APhysicalSite(s) ID(s) (FKs) - List of Transport Issue Log ID(s) (FK) - List of TransportRoutes ID(s) (Fk) - actual start datetime - actual end datetime	MH	High	DONE	

9	Transport	NF	The System must follow this format for TransportAreas: - ID -area name (string)	MH	High	DONE	
10	Transport	NF	For any TransportRoute in the System, a TransportRoute should have: - TransportRoute ID - Status(PENDING,STARTED,END..) -Source site ID (FK) -Destination site ID (FK) - Delivery File ID (FK)	MH	High	DONE	
11	Transport	NF	The System must send a Transport for dependent Suppliers only.	MH	High	DONE	

12	Transport	NF	In a given Transport the System must know from which suppliers to which branches the supplies should be distribued, via TransportRoute	MH	High	DONE	
13	Transport	F	The system must allow to check for requested items at supplier - if they exist - in which suppliers	NTH	Low	DONE	
14	Transport	F	The System must allow an admin to assign a user to the role of Transport Manager	NTH	Low	TBD	
14	Transport	F	The System must allow to define a user as admin	NTH	High	TBD	
15	Transport	F	The System must allow to define a user as transport manager	NTH	High	TBD	
16	Transport	F	The System must allow a transport manager/user to create a future transport	MH	High	DONE (SUPPORT ONLY USER)	
17	Transport	F	The System must allow a transport manager/user to update a transport	MH	High	DONE (SUPPORT ONLY USER)	
18	Transport	F	The System must allow a transport manager/user to delete a transport	MH	High	DONE (SUPPORT ONLY USER)	
19	Transport	F	The System must allow a transport manager/user a transport manager to create a driver	MH	High	DONE (SUPPORT ONLY USER)	

20	Transport	F	The System must allow a transport manager/user to edit a driver	MH	High	DONE (SUPPORT ONLY USER)	
21	Transport	F	The System must allow a transport manager/user to delete a driver	MH	High	DONE (SUPPORT ONLY USER)	
22	Transport	F	The System must allow a transport manager/user to create a truck	MH	High	DONE (SUPPORT ONLY USER)	
23	Transport	F	The System must allow a transport manager/user to edit a truck	MH	High	DONE (SUPPORT ONLY USER)	
24	Transport	F	The System must allow a transport manager/user to delete a truck	MH	High	DONE (SUPPORT ONLY USER)	
25	Transport	F	The System must allow a admin/user to create a branch	MH	High	DONE (SUPPORT ONLY USER)	
26	Transport	F	The System must allow a admin/user to edit a branch	MH	High	DONE (SUPPORT ONLY USER)	
27	Transport	F	The System must allow a admin/user to delete a branch	MH	High	DONE (SUPPORT ONLY USER)	
29	Transport	F	The System must allow a admin/user to create a TransportArea	MH	High	TBD	
30	Transport	F	The System must allow a admin/user to edit a TransportArea	MH	High	TBD	

31	Transport	F	The System must allow a admin/user to delete a TransportArea	MH	High	TBD	
32	Transport	F	During a given Transport - if a requested item(s) is missing from the relevant supplier - The System must allow the truck to be re-routed to a different supplier(s) to receive the missing item(s).	NTH	High	TBD	
33	Transport	F	When a Transport starts - the System must change the status of the Transport(status + datetime of the start), in all the Transport routes that connected to it.	NTH	High	DONE	
34	Transport	F	The System must alert in case that the initial truck weight is above the maximum truck weight	MH	Low	DONE	
35	Transport	F	Upon maximum truck weight overflow - the system will ask the TransportManager/user which strategy to use: - cancel or change one of the destinations - change truck - unload certain items from the current truck (Basically give the option to edit the entire transport)	MH	High	DONE (SUPPORT ONLY USER)	

36	Transport	F	The System can not allow a driver without a sufficent drivers license type to drive any truck	MH	High	DONE	
37	Transport	F	The System have to show the TransportArea by ID	MH	High	DONE	
38	Transport	F	The System have to show all the TransportAreas in the System	MH	High	DONE	
39	Transport	F	The System have to have show all the Transport in the System	MH	High	DONE	
40	Transport	F	The System should have show all the Transport by ID in the System	MH	High	DONE	
41	Transport	F	The System have to Print transport drives for driver - by Transport ID	NTH	High	DONE	
42	Transport	F	The System have to allow to Visit source of Transport Transport's route	NTH	High	DONE	
43	Transport	F	The System have to allow to destination source of Transport Transport's route	NTH	High	DONE	
44	Transport	F	The System have to show truck by its ID	MH	High	DONE	
45	Transport	F	The System have to show all the trucks in the System	MH	High	DONE	
46	Transport	F	The System have to show drivers by its ID	MH	High	DONE	
47	Transport	F	The System have to show all the drivers in the System	MH	High	DONE	
48	Transport	F	The System have to show branch by its ID	MH	High	DONE	

49	Transport	F	The System have to show all the branch in the System	MH	High	DONE	
50	Transport	F	The System have to show gui for handle all the models	MH	High	DONE	
51	Transport	NF	The System have to load/unload by the chioce of the user mock values (only for this version)	MH	High	DONE	
52	Transport	NF	The system must check if there is at least one available Stocker by date and time to start the Transport	MH	High	TBD	
53	Transport	NF	The system must check if there is at least one available Driver by time and date to start the Transport	MH	High	TBD	
54	Transport	NF	The system must synchornize with Employ system that drivers will be one to one with employ table in Employ system	MH	High	TBD	

ID	Module	Func/Non-Func	Description	Priority	Risk	Status
1	Stock	F	The system MUST give a notification before an item runs out of stock	MH	HIGH	TBD
2	Stock	F	The system MUST provide the option to choose the minimum quantity of the product before a notification is sent.	MH	LOW	DONE
3	Stock	NF	The system MUST follow this format for each product info: product ID amount, where, manufacturer, amount on shelves, amount in the warehouse.	MH	LOW	DONE
4	Stock	F	The system MUST be able to generate stock status for all the products in the store	MH	HIGH	DONE
5	Stock	NF	the system MUST follow this format for category: product ID, purchase price, amount in warehouse, amount in store shelves, sells price.	MH	LOW	DONE
6	Stock	NF	the System MUST follow the format for damaged products: BatchID, product ID, LocationType(warehouse,store shelves).	MH	LOW	DONE

			amount.			
7	Stock	F	The system MUST be able to make a report based on a category or categories	MH	HIGH	DONE
8	Stock	F	The system MUST be able to make a report of all defective items	MH	HIGH	DONE
9	Stock	F	The system MUST have the option update the amount of the products	MH	Low	DONE
10	Stock	F	The System MUST have the option to add a new product to the stock	MH	HIGH	DONE
11	Stock	F	The System MUST have the option to put to sale a category	MH	HIGH	DONE
12	Stock	F	The System MUST return all the products of a same category	MH	HIGH	TBD
13	Stock	F	the System MUST generate for every product a unique id	MH	llow	DONE
14	Stock	F	the System MUST generate for each category a unique id.	MH	low	DONE
15	Stock	F	the system MUST sort the products into three category each subcategory of the other	MH	HIGH	DONE
16	Stock	F	the System MUST to have the option to apply a discount on product or category(not both)	MH	HIGH	DONE
17	Stock	F	the system MUST track the product expiration date	MH	HIGH	DONE
18	Stock	F	the SHOULD support of moving product from the warehouse to the store shelve	NTH	LOW	DONE
19	Suppliers	F	the system must support adding new suppliers and saving their information in the format:			Done

			name , address , phone , number (unique and not null) , payment method , payment time , bank details and contact people which is a list of <name , phone number>			
20	Suppliers	F	the system will allow updating the details of existing suppliers in the system (the details are mentioned in 19)			Done
21	Suppliers	F	the system will allow deleting suppliers from the system			Done
22	Suppliers	F	the system will support creating orders from suppliers in the system , an order will contain the following information: supplier information , address , date , phone , status, list of the products in the order			Done
23	Suppliers	F	the system will allow updating the details of existing orders if they are in a non delivered status			Yet to start
24	Suppliers	F	the system will allow deletion of orders			Done
25	Suppliers	F	the system will keep track of status of orders and allow the user to change the status of the orders as he likes			Done

26	Suppliers	F	the system will support the functionality of showing orders by their status and in general all the orders as the user likes , tin this functionality the system will show the orders in a shorter form: supplier info, address , date , phone , id, status			Done
27	Suppliers	F	the system will allow showing the full details of an order (including the list of items,quantity as well as price descriptions) by id			Done
28	Suppliers	F	the system will support showing all the suppliers in the system in a shortened form as follows: name , address , supplier's number , payment method , payment time , bank details			Done
29	Suppliers	F	the system will allow showing the full details of a supplier by his id , the details as mentioned in the requirement 19			Done
30	Suppliers	F	the system should support searching suppliers by different categories : name,number			Done
31	Suppliers	F	the system should support searching orders by different categories : by id			Done

32	Suppliers	F	the system should make the range of categories which by the users can search orders and suppliers as big as possible (by every field of supplier/order)			Yet to start
33	Suppliers	F	the system should support combination of search categories , for example , the users can search suppliers by names and address			Yet to start
34	Suppliers	F	the system must support assigning contracts to suppliers in the system , a contract should have the following information : supplier information , list of contract terms contract term contains a product's information , price and an amount specifier amount specifier contains information regarding discount per quantity , so minimal quantity to receive the discount , base discount percentage			Done
35	Suppliers	F	the system should support adding amount specifiers to contracts , an amount specifier is explained in 34			Done
36	Suppliers	F	the system should support editing existing amount specifiers which are			Yet to start

			already attached to contract terms , explained in 34			
37	Suppliers	F	the system should support removal of existing amount specifiers which are already attached to contract terms , explained in 34			Yet to start
38	Suppliers	F	the system should allow updating a contract details , as explained in 34			Yet to start
39	Suppliers	F	the system will allow deletion of existing contracts as explained in 34			Yet to start
40	Suppliers	F	the system must allow the user to view history of purchases as well as meta data about the overall history : total sum , number of products purchased			Done
41	Suppliers	NF	the system must respond within 10 seconds for each user interaction with probability of 99%			Done
42	Suppliers	NF	the system should be simple to use with clear user interfaces and clear options			Done
43	Suppliers	NF	the system should save important information : orders,suppliers,contracts			Done

44	Suppliers	NF	the system must save it’s state from previous interactions			Yet to start
45	Suppliers	NF	the system should be well-oriented to allow users to reach any wanted page in very few interactions (2 - 4) with clear paths			Done
46	Suppliers& Stock		the system MUST have the option to add a new periodic order which have information as normal orders as explained in 22 , as well as delivery day			TBD
47	Suppliers& Stock		the system must allow the user to trigger automatic creation of orders which should make every product in the stock about it’s own critical level .			TBD
48	Suppliers& Stock		the system will support updating periodic orders however it will limit the time for updating to one day before the date of the delivery of the order .			TBD
49	Suppliers& Stock		the system will support showing a list of all products which are under their critical level , the products will be shown in a shorten form : product id , current quantity and critical level			TBD

דרישות פתוחות:

#	Topic	Issue
1	מחיקה של ספקים	אפשר לעדכן או למחוק ספקים מהמערכת ?
2	מספק לא קיים	אפשר הזמנה מספק שלא קיים במערכת ? אם כן אז איך תתנהג ת במצב כזה ?
3	הזמנות	אפשר עדכון הזמנות אחרי שכבר נוצרו ? אם כן אז האם יש תנאים זו ?
4	הזמנות	מערכת צריכה לעקוב אחרי סטטוס של כל הזמנה ? אם כן אז האם למשתמש לעדכן אותה או רק על ידי המערכת או הרשאות מיוחדות ?
5	המערכת	צופה מהמערכת להגיב "במהירות" ? אם כן אז נא להגדיר סף
6	ויות בהסכמים	חייב צירוף כתבי כמויות רק בעת הכנסת הסכם למערכת ? או לנהל אותם לגמרי ?
7	ספקים	' המערכת לאפשר הצגה של כל הספקים ? אם כן אז איך בדיוק ?
8	הזמנות	' המערכת לתמוך בהצגת הזמנות ? אם כן אז איך בדיוק ?
9	לספקים	' המערכת לאפשר חיפושים לפי קטגוריות עבור ספקים ?
10	הזמנות	' המערכת לאפשר חיפוש הזמנות ומיונם/סינונם לפי קטגוריות ?
11	הסכמים	' המערכת לאפשר חיפוש הסכמים לפי קטגוריות ובנוסף לאפשר חיפוש ק/פריט מסוים ?

Does the system need to provide the option to add new stock? ans:the system should have the option to add new products.	add new stock	12
Does the management want the system to calculate the sale or only to give the information about a specific product?	In the store, various sales can be managed for different products.	13
the report should present the total amount of products under same category or for each product under the category show the total amount? ans:the report presents the total amount in each category.	category report	14
witch discount applies if you have two discount on a product? ans: the discount with the most up to date start date will be preferable	Discount	15
Does the program need to keep a list of all the discounts? ans: no	Discount	16
Does the program need to search products by their name or ID? ans: ID	Products search	17
In the category stock report should the report should the stock per category and count each item once or for each category with no repeating items? ans: no repeating items	Category report	18
If a product is damaged should it be counted as an existing item to know if we need to buy more from the supplier?	Item counter	19

ans: should not be counted		
If an item is low in stock how much do we need to order? ans: order an item to the low stock quantity	order stock	20

#	Topic	Issue	
1.	Employment Contract	What are the employment conditions?	Sick Days, Vacation Days, Pension, Min Working hours per month,
2.	Salary	Global salary or Hourly rate ?	Hourly
3.	Numebr of Store	How many stores are opening	For now only One.
4.	SuperMarket Roles	What Roles are there in the SuperMarket?	Cashier , Manger , Stocker , Cleaner , Security , HR Lead
5.	Permissions Hierarchy	Who can Accept / Terminate employees?	Only HR Lead

<u>שינויים בטבלאות דרישות מודל העובדים מאז חלק א:</u>

1) בעקבות ההערות מחלק א של העבודה הבנו שדרישות 11, 16 היו מנוסחות בצורה לא מדויקת ולכן דייקנו את הניסוח.
   בנוסף הבנו ששתי הדרישות הללו אמורות להיות פונקציונליות ולא NF כי הדרישות הללו מתאימו יותר להגדרה של דרישה פונקציונלית מאשר דרישה לא פונקציונלית.

2) דרישה מס 8 הייתה מנוסחת בצורה לא טובה כי לא דייקה את הדרישה שהתכוונו אליה לכן כעת ניסחנו את הדרישה שתתאים למימוש אנחנו מתכוונים לעשות בקוד שלנו.

3) הוספנו את דרישות 21-27 על מנת לאפשר אינטגרציה (שילוב) חלקה עם מודל ההובלות.

<u>שינויים בטבלת ספקים ומחסן:</u>

1)צירפנו את הטבלה של שאר המודלים(עובדים והובלות)

2) שינינו את טבלת הספקים כך שהיא תתאים לפורמט של שאר הטבלאות(באנגלית)

3)הוספנו את 46-49 בשביל שיהיה איחוד וקישור בין המודלים מחסן וספקים

<u>הגדרות עבור מודל העובדים:</u>

מנהל כוח אדם (HR):
עובד בעל הרשאות ניהול גבוהות, אחראי על הוספה והסרה של עובדים מהמערכת, קביעת שעות משמרת, הגדרת הדרישות לסוגי התפקידים במשמרת והכמות מכל אחד מהם.

מנהל משמרת:
עובד עם הרשאות ניהול משמרות, יכול ליצור משמרות, לשבץ ולהסיר עובדים מהן, לקדם עובדים לתפקידים נוספים ולבצע פעולות ניהוליות הקשורות למשמרות בלבד.

עובד כללי :
משתמש במערכת ללא הרשאות ניהול, יכול לעדכן זמינות למשמרות בשבוע הקרוב, לצפות בפרטיו האישיים ולבצע פעולות בסיסיות הקשורות לעבודתו בלבד.
כרגע העובדים הכללים במערכת הם: קופאי\ת, מחסנאי\ת, מנקה ונהג.

זמינות :
היכולת של עובד לבחור או לעדכן באילו ימים או שעות הוא פנוי לעבוד, לצורך שיבוץ למשמרות.
עידכון הזמינות מתאפשר לעשות רק עד שבוע קדימה.

חוזה :
המידע הבסיסי שמגדיר את תפקידו של העובד בארגון, כמות ימי החופשות, אחוז הפנסייה, כמות שעות עבודה מינימלית בחודש ועוד.

תפקיד :
במערכת שלנו קיימים 7 תפקידים- קופאי, מנקה, מחסנאי, מנהל משמרת, מנהל כוח אדם, נהג ומאבטח.

<u>הגדרות עבור מודל הובלות:</u>

מנהל מערכת - אחראי על הגדרת מנהל הובלה (Administer)

מנהל הובלה - אחראי על יצור של הובלות חדשות, מחיקה של הובלה קיימת, עדכון של הובלה קיימת

בנוסף אחראי על סידור נהגים ומשאיות לפי כל הובלה

מסלול הובלה - מסלול הובלה מספק אחד אל יעד אחד על ידי נהג יחיד ומשאית יחידה. בדיקה בכל מסלול אם המשאית עומדת במשקל ואם לא חסרים פריטים אשר אמורים להגיע ליעד. בנוסף, בדיקה אם קיים מחסנאי זמין ביעד לסדר את הסחורה.

הובלה - אוסף של מסלולי הובלה כאשר מצוות לכל המסלולים נהג אחד ומשאית אחת, בדיקה שכל התנאים מתקיימים וניתן להוציא הובלה לפועל.

נהג - עובד אשר הוגדר כנהג לפי רישיון מתאים וכדומה

משאית - כלי רכב שהוגדר לפי משקל, סוג רישיון וכדומה.

<u>הגדרות מודל מחסן:</u>

product - מוצר אשר נימצא במחסן בעל מספר מזהה,שם, יצרן,וכמה יש ממנו בכללי בתוך החנות.

productBatch- כמות מסויימת של מוצר מסויים מאופיין על ידי מס מזהה באצ', מוצר, כמות מאותו המוצר והיא הוא נימצא וגם תאריך תפוגה שלו ואם המוצר פגום או לא. כלומר למוצר יכל להיות הרבה באצ'ים.

productCategory- לכל מוצר מקוטלג על פי שלושה קטגוריות כך שכל קטגוריה היא תת קטגוריה של השנייה.

discounts - הנחה על מוצר מסויים או קטגוריה אך לא על שניהם ביחד מאופיין עם תיאור קטן על הנחה כמות אחוזים וטווח תאריכים לאותה הנחה.

Full instructions guide for Employees and Transport systems.

The ultimate guide for transport and employees' systems.

Denis Coleman, Yael Durahly, Ethen Krimer, Tom Gluzman

Names and IDs:

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Welcome to Employees Management system Instructions!!

*Note: instructions are meant for the customer/user.

Summary:

The following employee management system provides a comprehensive framework for managing workforce operations, ensuring efficient and secure access for different roles. The system starts with a Welcome Panel, where users enter their unique ID to gain personalized access, reinforcing security and role-based functionality.

Once inside, regular employees can access the General Employee Panel, which allows them to update their shift availability, view personal information, and manage basic work-related actions. This panel serves as the foundation for all employees, ensuring they can quickly check and update their status within the organization.

For employees with managerial roles, the system offers a specialized Manager Panel, which expands their capabilities to include creating new shifts, adding or removing employees from shifts, promoting employees, and using cancel cards. Managers also have the flexibility to return to the general panel as needed, providing seamless navigation between different levels of authority.

Additionally, the system includes a dedicated HR Panel for those responsible for higher-level organizational control. This panel includes critical actions such as terminating or adding employees, setting global shift hours, and defining required roles for each shift. It allows HR personnel to update role requirements and manage the overall structure of shift assignments, ensuring the workforce remains balanced and efficient.

Together, these panels form a cohesive system that supports the diverse needs of regular employees, managers, and HR personnel, while maintaining detailed historical data for accurate oversight and reporting.

How to Use the System:

When you start the system, you will first decide whether to use the provided mock data or start with a clean database.

```
Load mock data?
[Warning!!! This will wipe(erase) the current DB, the point of the mock data is to give you a minimal working fresh DB]
(y/n):
```

Once this choice is made, you will be presented with a welcome panel displaying both the Employee and Transport panels, where you can choose which one to enter or to exit the system.

```
Welcome to the Transport+Employee Management System!
Please choose an option:
1. Transport system
2. Employee system
3. Exit
```

If you choose to enter the Employee Panel, you will be greeted by the Login Panel, where you can either enter your unique employee ID to access your personalized interface or choose to exit, which will return you to the previous welcome panel.

```
Welcome to the Employees Management System!

You are on Login panel!


Please choose an action (1 or 2):
1. Enter to your Employee Role
2. Exit System
```

After a successful login, you will gain access to your personalized **General Employee Interface**, where you can perform the following actions.

1. Set Upcoming Shifts Availability – Allows you to indicate which days and times you are available to work in the upcoming week. This helps shift managers assign you to suitable shifts.

2. Enter to your Managing Role – If you hold a management role (such as Shift Manager or HR), this option takes you to the appropriate management panel, where you can perform role-specific administrative tasks.

3. Show Your Personal Information – Displays your stored personal details, including your name, ID, role(s), and any relevant employment information currently in the system.

4. Show all Shifts for the next week – Presents a schedule of all upcoming shifts across the system, allowing you to see which shifts are scheduled and prepare accordingly.

5. Going back to Login panel – Logs you out of your session and returns you to the login screen, where another user can enter or the system can be exited.

```
Welcome to the General Employee Panel!

Please choose an action (1 to 5):
1. Set Upcoming Shifts Availability
2. Enter to your Managing Role
3. Show Your Personal Information
4. Show all Shifts for the next week
5. Going back to Login panel
```

The **HR Panel** provides advanced administrative controls for managing the employee database and shift structure.
Upon entering this panel, HR personnel are presented with the following options:

1. **Terminate Employee from System:** The system will prompt the HR lead to enter the necessary employee information to remove them from the database, permanently deleting their records.
2. **Add New Employee to System:** HR can onboard new employees by providing the required personal and role details, adding them to the active employee database.
3. **Set Default Shift Hours:** HR can define the default shift hours for all future shifts.
4. **Update Required Roles for a Shift:** This option allows HR to adjust the required number of employees for a specific role for all the shifts.
5. **Set All Required Shift Roles:** HR can bulk update the staffing requirements for all roles, setting the default number of employees needed for each role across all shifts.
6. **Update Employee Information:** This feature allows HR personnel to modify existing employee records in the system. It includes updating personal details, role assignments, availability schedules and more according to the entered choice.

7. **Return to General Employee Panel:** HR users can exit the HR panel and return to the general employee interface, providing flexibility to switch between administrative and regular functions.

```
Welcome to the HR Panel!

Please choose an action (1,2,3,4,5,6 or 7):
1. Terminate Employee from System
2. Add New Employee to System
3. Set Shift Hours
4. Update Required Roles to Shift
5. Set All Required Shift Roles
6. Update Employee's information
7. Going back to General Employee panel
```

The **Shift Manager Panel** offers a range of tools for shift management and employee role adjustments.
Upon accessing this panel, managers are presented with the following options:

1. **Create New Upcoming Shift:** The manager can create new upcoming shifts by utilizing the system's available tools to define shift details and requirements.
2. **Remove Employee from Upcoming Shift:** Allows the manager to remove an employee from a scheduled upcoming shift as needed.
3. **Add Employee to Upcoming Shift:** The manager enters the employee ID, shift ID, and the desired role to assign the employee to that specific role in the shift. The system then verifies the employee's availability and confirms that the employee holds the requested role before completing the assignment.
4. **Promote Employee:** The manager can assign an additional role to an employee, expanding their responsibilities and permissions within the system.
5. **Cancel Items from Shopping Cart:** Using their manager card, the manager can cancel selected items from the shopping cart when necessary.
6. **Return to General Employee Panel:** Provides the option to exit the Managing Panel and return to the General Employee Panel.

```
Welcome to the Manager Panel!

Please choose an action (1,2,3,4,5 or 6):
1. create New Shift
2. Remove Employee from Shift
3. Add Employee to Shift
4. Promote Employee
5. Use cancel card
6. Go back to General Employee panel
```

At each step, if the user enters incorrect information, the system will notify him and prompt him to provide the correct details, ensuring a smooth and secure user experience.

Database:

The Employee Management System utilizes two mock databases for efficient data handling.

The **Employee Database** stores all the essential information about each employee, including their unique ID, name, list of roles, contract, salary, recruitment date, bank information and availability.

The **Shift Database** is divided into two sections: **Past Shifts** and **Upcoming Shifts**. Shifts are automatically transferred to the **Past Shifts** section once their scheduled time has passed, ensuring accurate tracking of work history.

For demonstration purposes, the mock system includes a sample database with 5 employees and 3 shifts, providing a realistic test environment for the system's functionality.

Employees Database information:

Employee 1

- Employee ID: EMP001
- Name: Alice
- Hourly Wage: 32.5

- Start Date: 15/03/2022
- Roles: Stocker
- Bank Account: Leumi-Hills (Acc: 072543, Branch: 972)
- Contract: 7 hours/day, 14 days/month, 21 shifts/month, 160 hours/month
- Availability:
    - Sunday: Day
    - Monday: Day
    - Tuesday: Day
    - Wednesday: Night
    - Thursday: Unavailable

Employee 2

- Employee ID: EMP002
- Name: Bob
- Hourly Wage: 47.1
- Start Date: 10/01/2023
- Roles: Shift Manager, Stocker
- Bank Account: Hapoalim (Acc: 072923, Branch: 001)
- Contract: 5 hours/day, 10 days/month, 14 shifts/month, 80 hours/month
- Availability:
    - Sunday: Day
    - Monday: Night
    - Tuesday: Night
    - Wednesday: Day
    - Thursday: Day

Employee 3

- Employee ID: EMP003
- Name: Charlie
- Hourly Wage: 120.2
- Start Date: 20/05/2021
- Roles: HR Lead
- Bank Account: Bank-Yahav (Acc: 072577, Branch: 625)
- Contract: 6 hours/day, 12 days/month, 18 shifts/month, 120 hours/month
- Availability:
    - Sunday: Day
    - Monday: Night
    - Tuesday: Day
    - Wednesday: Night
    - Thursday: Day

Employee 4

- Employee ID: EMP004
- Name: David
- Hourly Wage: 35.0
- Start Date: 01/06/2023
- Roles: Stocker
- Bank Account: Bank-Hapoalim (Acc: 072589, Branch: 311)
- Contract: 5 hours/day, 10 days/month, 14 shifts/month, 80 hours/month
- Availability:
    - Sunday: Day
    - Monday: Day

    - Tuesday: Day
    - Wednesday: Day
    - Thursday: Night

Employee 5

- Employee ID: EMP005
- Name: Eve
- Hourly Wage: 33.5
- Start Date: 15/07/2023
- Roles: Stocker, Cleaner
- Bank Account: Discount (Acc: 072111, Branch: 321)
- Contract: 6 hours/day, 12 days/month, 18 shifts/month, 120 hours/month
- Availability:
    - Sunday: Both
    - Monday: Both
    - Tuesday: Both
    - Wednesday: Both
    - Thursday: Both

Employee 6

- Employee ID: EMP006
- Name: Inna
- Hourly Wage: 33.0
- Start Date: 10/07/2024
- Roles: Stocker, Security
- Bank Account: Bank-Yahav (Acc: 0725431, Branch: 972)

- Contract: 6 hours/day, 12 days/month, 18 shifts/month, 120 hours/month
- Availability:
    - Sunday: Both
    - Monday: Both
    - Tuesday: Both
    - Wednesday: Both
    - Thursday: Both

Employee 7

- Employee ID: EMP007
- Name: John
- Hourly Wage: 33.0
- Start Date: 10/07/2024
- Roles: Driver
- Bank Account: Bank-Yahav (Acc: 0725431, Branch: 972)
- Contract: 6 hours/day, 12 days/month, 18 shifts/month, 120 hours/month
- Availability:
    - Sunday: Both
    - Monday: Both
    - Tuesday: Both
    - Wednesday: Both
    - Thursday: Both

Employee 8

- Employee ID: EMP008
- Name: Sam
- Hourly Wage: 33.0
- Start Date: 10/07/2024
- Roles: Driver
- Bank Account: Bank-Yahav (Acc: 0725431, Branch: 972)
- Contract: 6 hours/day, 12 days/month, 18 shifts/month, 120 hours/month
- Availability:
    - Sunday: Both
    - Monday: Both
    - Tuesday: Both
    - Wednesday: Both
    - Thursday: Both

Employee 9

- Employee ID: EMP009
- Name: Jane
- Hourly Wage: 33.0
- Start Date: 10/07/2024
- Roles: Driver
- Bank Account: Bank-Yahav (Acc: 0725431, Branch: 972)
- Contract: 6 hours/day, 12 days/month, 18 shifts/month, 120 hours/month
- Availability:
    - Sunday: Both
    - Monday: Both

- Tuesday: Both
- Wednesday: Both
- Thursday: Both

Shifts Database information:

Shift 1 – past shift

• Shift ID: SHIFT-1

• Shift Type: Day

• Date: 15/05/2021

• Manager: Alice (Employee ID: EMP001)

• Assigned Employees:

Alice (Employee ID: EMP001) - (Role: Stocker)

Bob (Employee ID: EMP002) - (Role: Shift Manager)

- Available Employees:
  Inna (Employee ID: EMP006) – (Role: Stocker)

Shift 2 – past shift

• Shift ID: SHIFT-2

• Shift Type: Night

• Date: 12/08/2016

• Manager: Charlie (Employee ID: EMP003)

• Assigned Employees:

Charlie (Employee ID: EMP003) - (Role: HR Lead)

David (Employee ID: EMP004) - (Role: Stocker)

• Available Employees:

Alice (Employee ID: EMP001) - (Role: Stocker)

Shift 3 – past shift

• Shift ID: SHIFT-3

• Shift Type: Day

• Date: 27/05/2025

• Manager: Bob (Employee ID: EMP002)

• Assigned Employees:

Bob (Employee ID: EMP002) - (Role: Shift Manager)

Eve (Employee ID: EMP005) - (Role: Cleaner)

John (Employee ID: EMP007) - (Role: Driver)

• Available Employees:

Charlie (Employee ID: EMP003) - (Role: HR Lead)

Shift 4 - future

• Shift ID: SHIFT-4

• Shift Type: Night

• Date: 26/05/2025

• Manager: Bob (Employee ID: EMP002)

• Assigned Employees:

Bob (Employee ID: EMP002) - (Role: Shift Manager)

David (Employee ID: EMP004) - (Role: Stocker)

Sam (Employee ID: EMP008) - (Role: Driver)

• Available Employees:

John (Employee ID: EMP002) - (Role: Driver)

Shift 5 - future

• Shift ID: SHIFT-5

• Shift Type: Day

• Date: 30/05/2025

• Manager: Bob (Employee ID: EMP002)

• Assigned Employees:

Bob (Employee ID: EMP002) - (Role: Shift Manager)

Alice (Employee ID: EMP001) - (Role: Stocker)

Jane (Employee ID: EMP009) - (Role: Driver)

• Available Employees:

Eve (Employee ID: EMP005) - (Role: Stocker)

For Your Information:

- The system handles HR Lead options permanently, meaning every change he does will be affective from this point forward until he will decide otherwise.
  For example setting shift hours will be the ones that he decides for all shifts made from this point onward.
- The system holds Past Shifts up to 7 years, then deletes them.
- The system accommodates fully the User throughout his use of the system, no user will be left out. The system gives explanations about each step and has fail safes.

Welcome to Transport Management system Instructions!!

*Note: instructions are meant for the customer/user.

Login Information

By default, the system does not require a login. If access control is introduced, usernames and passwords must be documented here.

System Purpose

This system is a command-line Java application designed to help logistics managers efficiently create, monitor, and manage transports between suppliers and branches using drivers and trucks. It includes full CRUD (Create, Read, Update, Delete) capabilities for all major components such as branches, drivers, trucks, and transport routes.

System Launch Instructions

1. Open a terminal window.

2. Navigate to the project directory.

3. Run the system using the following command:

`java -jar release/adss2025_v01.jar`

4. The application starts with a prompt: 'Load mock data? (y/n)'.

- Type 'y' to initialize mock data (sample branches, trucks, drivers, and suppliers).

- Type 'n' to start from scratch.

Main Menu Walkthrough

The main menu includes:

- Manage Transport Areas: Connect suppliers and branches into transport zones.
- Manage Transports: Create, view, and delete logistics operations.
- Manage Trucks: Maintain vehicle records.
- Manage Drivers: Add or update drivers and their license types.
- Manage Branches: Track branch locations and contact info.
- Exit: Close the program safely.
- Load Temporary Mock Data: Reinitialize demo data at any point.

Creating a Transport

To create a transport:

1. Choose 'Manage Transports' from the main menu.
2. Select 'Create transport'.
3. Enter a planned date and time for delivery.
4. Choose a source branch.
5. Select items from inventory to include in the transport.
6. System will attempt to find routes and available trucks/drivers.
7. You will be prompted to pick from valid trucks and drivers.
8. Upon success, a full transport is created and listed.

Drivers and Trucks

Each driver must be assigned a license type (B, C, C1, etc.).

Each truck requires a specific license type to operate.

The system only allows pairing a driver with a truck they are qualified to drive.

Mock Data Description

The mock data includes:

Branches

Three branches are added:

- **Branch1** – Address: "Address1", Phone: "+972 5412311"
- **Branch2** – Address: "Address2", Phone: "+972 5488899"
- **Branch3** – Address: "Address3", Phone: "+972 5412988"

Drivers

Three drivers with different license types:

- **John Doe** – ID: "123456789", Phone: "+972 5412311", License: "C"
- **Sam Wilson** – ID: "987654321", Phone: "+972 5488899", License: "B"
- **Jane Smith** – ID: "456789123", Phone: "+972 5412988", License: "C1"

Trucks

Three trucks with different capacities and required license types:

- **Truck 5498875** – Weight: 3500kg, Max load: 5000kg, License: "C1", Manufacturer: "Toyota"
- **Truck 5491176** – Weight: 1000kg, Max load: 3000kg, License: "C", Manufacturer: "Ford"
- **Truck 5491177** – Weight: 4500kg, Max load: 7000kg, License: "C1", Manufacturer: "Mercedes"

Suppliers

Three suppliers located at different addresses:

- **Supplier1** – Address: "Address1", Phone: "+972 77788855"
- **Supplier2** – Address: "Address2", Phone: "+972 77788856"
- **Supplier3** – Address: "Address3", Phone: "+972 77788857"

Transport Areas

Two transport areas are created, each associating existing branches and suppliers:

- **Area1** – Includes all created branches and suppliers
- **Area2** – Includes a subset (first two) of branches and suppliers

Transport Planning

- A set of stock items is retrieved based on a mock search (with placeholder query "").
- A **transport route** is planned for the next day, using the selected stock items.
- The system finds an available truck and a driver with a matching license for the planned date.
- Finally, a **transport** is created linking the planned time, selected truck and driver, and the created transport routes.

Error Handling and Tips

The program uses Helper methods to validate user input. If an entry is invalid, the user is prompted again.

- Do not leave inputs blank.

- Follow format for dates and numbers.

- Ensure there are available drivers and trucks for the given transport date.

Extending the System

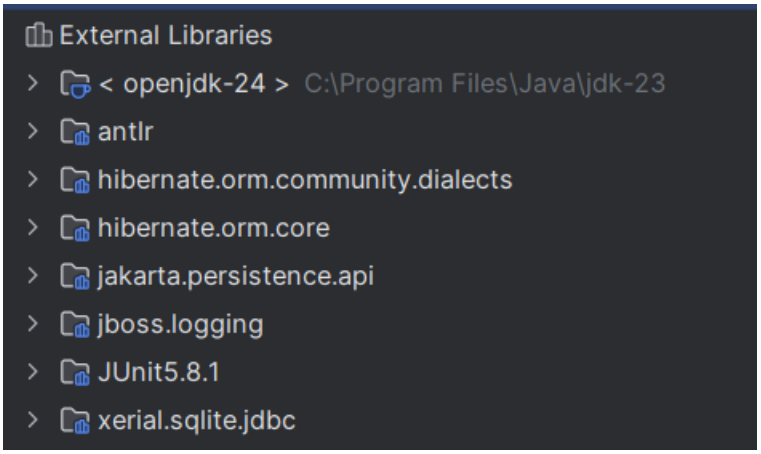
You can extend functionality by:

- Adding transport history and tracking.

- Introducing login/authentication features.

- Adding GUI components for ease of use

External libraries:



antlr – Used for parsing and analyzing structured text (like SQL or code), used by Hibernate for interpreting query languages.

hibernate.orm.community.dialects – Contains additional SQL dialects for supporting various databases in Hibernate.

hibernate.orm.core – The main Hibernate ORM framework, enabling mapping Java classes to database tables and managing persistent data.

jakarta.persistence.api – Provides the standard JPA interfaces (like Entity, EntityManager, etc.) which Hibernate implements.

jboss.logging – A logging framework used by Hibernate and related libraries for consistent, configurable logging.

JUnit 5.8.1 – A testing framework for writing and running unit tests in Java.

xerial.sqlite.jdbc – A JDBC driver that allows Java applications to connect to SQLite databases.

Good Luck!